

EXECUTIVE PROFILE

Chartered civil engineer with a 48-year career spanning the full arc of geospatial and design technology — from manual topographic survey and ink-on-mylar drafting, through the first wave of CAD adoption in the 1980s, the GIS revolution of the 1990s, and the transition to parametric BIM. Former technical director of a national infrastructure consultancy; now an independent advisor to engineering firms, public authorities and technology ventures. Deep, hands-on command of both the geospatial (QGIS, ArcGIS) and CAD/AEC (Civil 3D, Revit, FreeCAD, Rhino, Inventor) toolchains, with a rare ability to articulate the **reasoning** behind professional design decisions — making him an ideal contributor to AI research that seeks to capture genuine domain expertise.

48

YEARS IN PRACTICE

220+

PROJECTS DIRECTED

€1.4B

WORKS ENGINEERED

30+

YRS CAD & GIS

CURRENT & ADVISORY ENGAGEMENTS

Founder & Principal Consultant

2016 – Present

Licini Advisory — Independent Geospatial & BIM Practice, Milan

- Advises engineering firms and municipalities on GIS-to-BIM digital-twin strategy, data standards (IFC, CityGML) and tool selection across QGIS, ArcGIS Pro, Civil 3D and Revit.
- Engaged as subject-matter expert and curriculum reviewer for two European EdTech / AI ventures building professional-grade geospatial and CAD training corpora.
- Court-appointed technical expert (CTU) on infrastructure and cadastral disputes for the Tribunale di Milano.

Senior Technical Advisor (part-time)

2018 – Present

Lombardi Infrastructure Engineering S.r.l. — Milan

- Mentors BIM-coordination teams and reviews design models on major road and rail tenders; chairs internal quality-assurance for geospatial deliverables.

Adjunct Lecturer — Geomatics & BIM

2009 – Present

Politecnico di Milano — School of Civil, Environmental & Land Management Engineering

- Guest lectures on the evolution of CAD/GIS practice and on integrating survey data into BIM workflows; supervises master's theses bridging geomatics and parametric design.

CAREER HISTORY

Technical Director

1998 – 2016

Geotecnica Lombarda S.p.A. — Milan & Turin

- Directed a 60-strong engineering and survey division; introduced the firm's first enterprise GIS (ArcInfo → ArcGIS) and led the 2008–2012 migration from 2D CAD to full BIM (Revit / Civil 3D).
- Principal engineer on motorway, metro and flood-defence schemes totalling over €1.4 billion in constructed value.

Senior Design Engineer → Project Manager

1986 – 1998

Studio Ingegneria Riva & Partners — Turin

- Among the firm's first AutoCAD adopters (R2.5, 1987); built CAD-standards and layer conventions still echoed in today's templates. Delivered urban road, drainage and land-development designs.

Surveyor & Draftsman → Junior Engineer

1977 – 1986

Ufficio Tecnico Provinciale / Impresa Costruzioni Bergamasca

- Field topographic survey (theodolite, levelling), cadastral mapping and manual drafting on the board — the foundation that informs every digital workflow since.

TOOL PROFICIENCY & EXPERIENCE

QGIS		14 yrs	Rhino 3D		9 yrs
ArcGIS Pro / ArcMap		26 yrs	Autodesk Inventor		6 yrs
AutoCAD / Civil 3D		30+ yrs	MapInfo Pro		18 yrs
Autodesk Revit		16 yrs	FME · PostGIS · SQL		15 yrs
FreeCAD BIM / IFC		7 yrs	OpenSCAD / SolveSpace		3 yrs

EDUCATION & CHARTERSHIP

Laurea in Ingegneria Civile (5-yr, Old Ordinance)

Politecnico di Milano — 1977 · 110/110 e lode

Specialisation: Hydraulics, Transport & Topographic Surveying (Geomatica)

Chartered Engineer — Ordine degli Ingegneri della Provincia di Milano

Registered Sector A (Civil & Environmental) · n. 11427 · since 1978

Executive Programme — Geographic Information Systems for Infrastructure

ITC / University of Twente (NL) — 1994

PROFESSIONAL AFFILIATIONS & STANDARDS WORK

- Member, buildingSMART Italy — openBIM / IFC working group
- Senior Member, AGI — Associazione Geometri / Geomatica Italiana
- Court Technical Expert (CTU), Tribunale di Milano
- Contributor, UNI 11337 (Italian BIM standard) review panel
- Past Vice-Chair, regional chapter, Ordine degli Ingegneri
- Reviewer, two EU geospatial-AI training initiatives

SELECTED PUBLICATIONS & LECTURES

- § Licini, M. (2014). “From Mylar to Model: Four Decades of Design-Tool Evolution in Italian Civil Practice.” *Quaderni di Ingegneria del Territorio*, 27(2).
- § Licini, M. & Riva, P. (2011). “GIS-to-BIM Data Continuity in Linear Infrastructure Projects.” *Proc. SIFET National Geomatics Conference*, Parma.
- § Invited keynote — “Why Expertise, Not Just Software, Builds Good Infrastructure.” Politecnico di Milano BIM Forum, 2019.

LANGUAGES & PERSONAL

- Italian — Native
- French — B2
- English — C1, professional working proficiency
- German — A2 (technical reading)

Actively practising and not retired — engaged on a flexible, advisory basis and available for remote consulting and expert-contributor roles. Driven by a conviction that the judgement accumulated over a career — the why behind every alignment, grade and model decision — is precisely the knowledge worth preserving for the next generation of intelligent design tools.